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9.6. Types of skeletal muscle contraction

There are two primary types of muscle contractions. They are **isotonic contraction** and **isometric contraction**. The types of contractions depend on the changes in the length and tension of the muscle fibres at the time of its contraction.

Isotonic contraction (iso- same, ton- weight/resistance)

In isotonic contraction the length of the muscle changes but the tension remains constant. Here, the force produced is unchanged. Example: lifting dumbbells and weightlifting.

Isometric contraction (iso- same, metric-distance)

In isometric contraction the length of the muscle does not change but the tension of the muscle changes. Here, the force produced is changed. Example: pushing against a wall, holding a heavy bag.

Types of skeletal muscle fibres

The muscle fibres can be classified on the basis of their rate of shortening, either fast or slow and the way in which they produce the ATP needed for contraction, either oxidative or glycolytic. Fibres containing myosin with high ATPase activity are classified as fast fibres and with lower ATPase activity are classified as slow fibres. Fibres that contain numerous mitochondria and have a high capacity for oxidative phosphorylation are classified as **oxidative fibres**. Such fibres depend on blood flow to deliver oxygen and nutrients to the muscles. The oxidative fibres are termed as **red muscle fibres**. Fibres that contain few mitochondria but possess a high concentration of glycolytic enzymes and large stores of glycogen are called **glycolytic fibres**. The lack of myoglobin gives pale colour to the fibres, so they are termed as **white muscle fibres**.

Skeletal muscle fibres are further classified into three types based on the above classification. They are slow – oxidative fibres, fast – oxidative fibres and fast – glycolytic fibres.

1. **Slow – oxidative fibres** have low rates of myosin ATP hydrolysis but have the ability to make large amounts of ATP. These fibres are used for prolonged, regular activity such as long distance swimming. Long – distance runners have a high proportion of these fibres in their leg muscles.
2. **Fast – oxidative fibres** have high myosin ATPase activity and can make large amounts of ATP. They are particularly suited for rapid actions.
3. **Fast – glycolytic fibres** have myosin ATPase activity but cannot make as much ATP as oxidative fibres, because their source of ATP is glycolysis. These fibres are best suited for rapid, intense actions, such as short sprint at maximum speed.

9.7 Properties of Skeletal Muscles

The four major properties of skeletal muscles are

Excitability/Irritability—The ability to respond or contract in response to chemical and electrical signals.

Contractility—The ability to shorten which causes movement of the structures to which the muscles are attached.